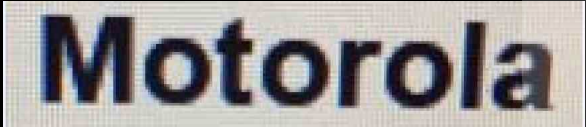
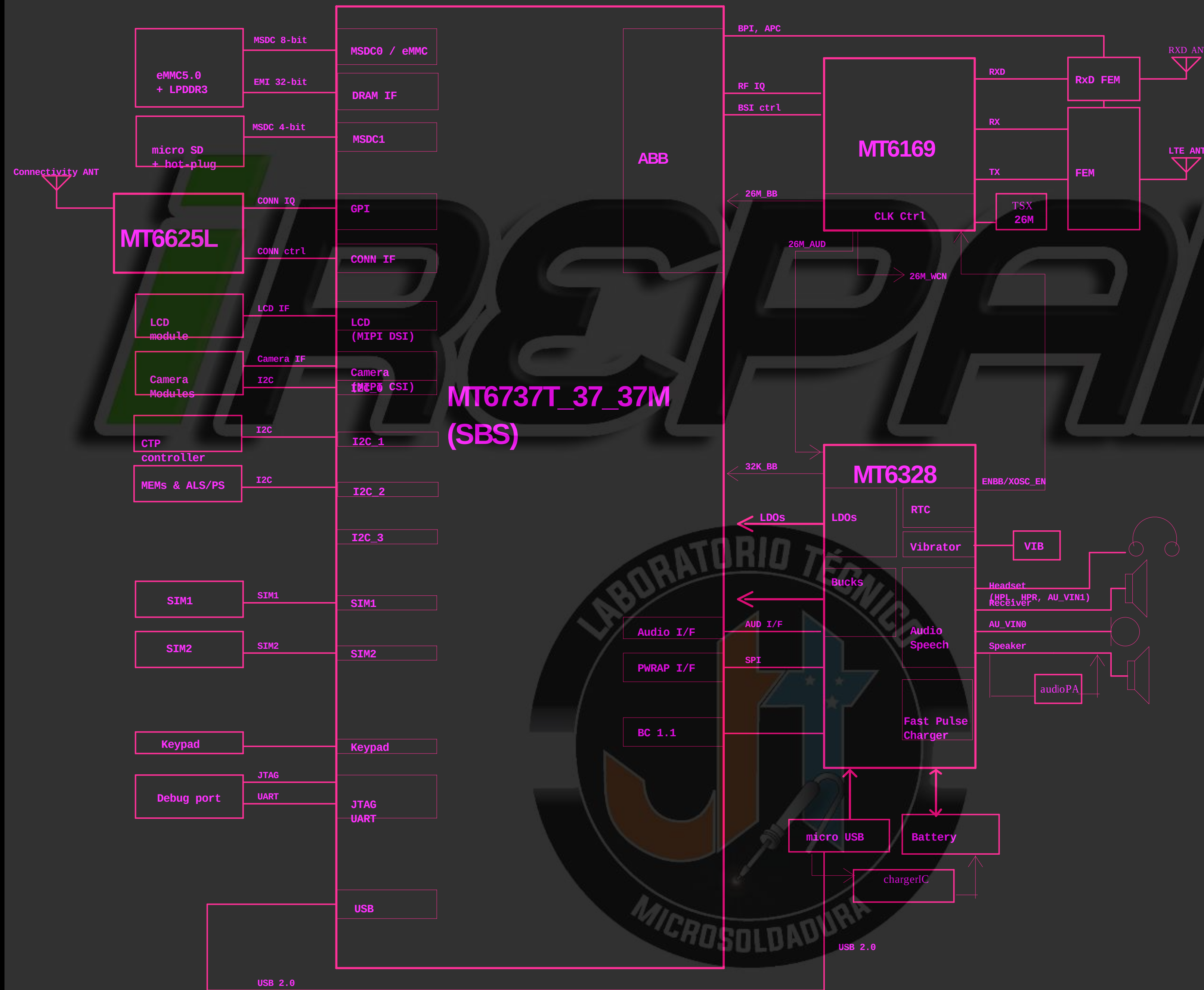


REVISION RECORD			
U/R	ECO NO.	APPROVED:	DATE:



XT1758 Block_diagram&SCH

Project : MT6737T_37_37M REF_SCH TOP LEVEL



COMPANY: Motorola			
TITLE: XT1758			
CODE: <Code>	SIZE: A1	DRAWING NO: <Drawing Number>	REV: 1_13
SCALE: <Scale>		SHEET: 21	21

DRAWN: <Drawn By>	DATED: <Drawn Date>
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: <Released By>	DATED: <Release Date>

REVISION RECORD			
UTR	ECO NO.	APPROVED	DATE

I2C	Function	I2C Spec.	Budget Timing	I2C Slave Address (7-bit mode)
I2C-0	Rear Camera - 8M	400 Kbps	Yes.	Rear camera (IMX135) I2C address: 0X10 (Write:0x20, Read:0x21) AF driver (DW9714A) I2C address: 0x0C (Write:0x18 , Read:0x19)
	Front Camera-2M	400 Kbps	Yes.	Front camera (6C2355) I2C address: 0x3C (Write:0x78, Read:0x79)
	charger IC			charger IC (FAN54015) I2C address: 0x6A (Write:0xD4, Read:0xD5)
I2C-1	CTP	400 Kbps	Yes.	GT1151 / CTP I2C address: 0X5D (Write:0xBA, Read:0xBB) or 0x14 (Write:0x28, Read:0x29)
I2C-2	LCD_DC-DC			DC-DC (KTD2151EUO-TR) I2C Address: 0x3E (Write:0x7C, Read:0x7D)
	G-Sensor	400 Kbps	Yes.	G-Sensor (BMA253) I2C Address: 0x18 (Write:0x30, Read:0x31)
I2C-3				
Note : I2C Spec. : Standard mode (100 kbps) and Fast mode (400 kbps), Fast mode Plus (1 Mbps) and High-speed mode (3.4 Mbps)				



COMPANY: Motorola

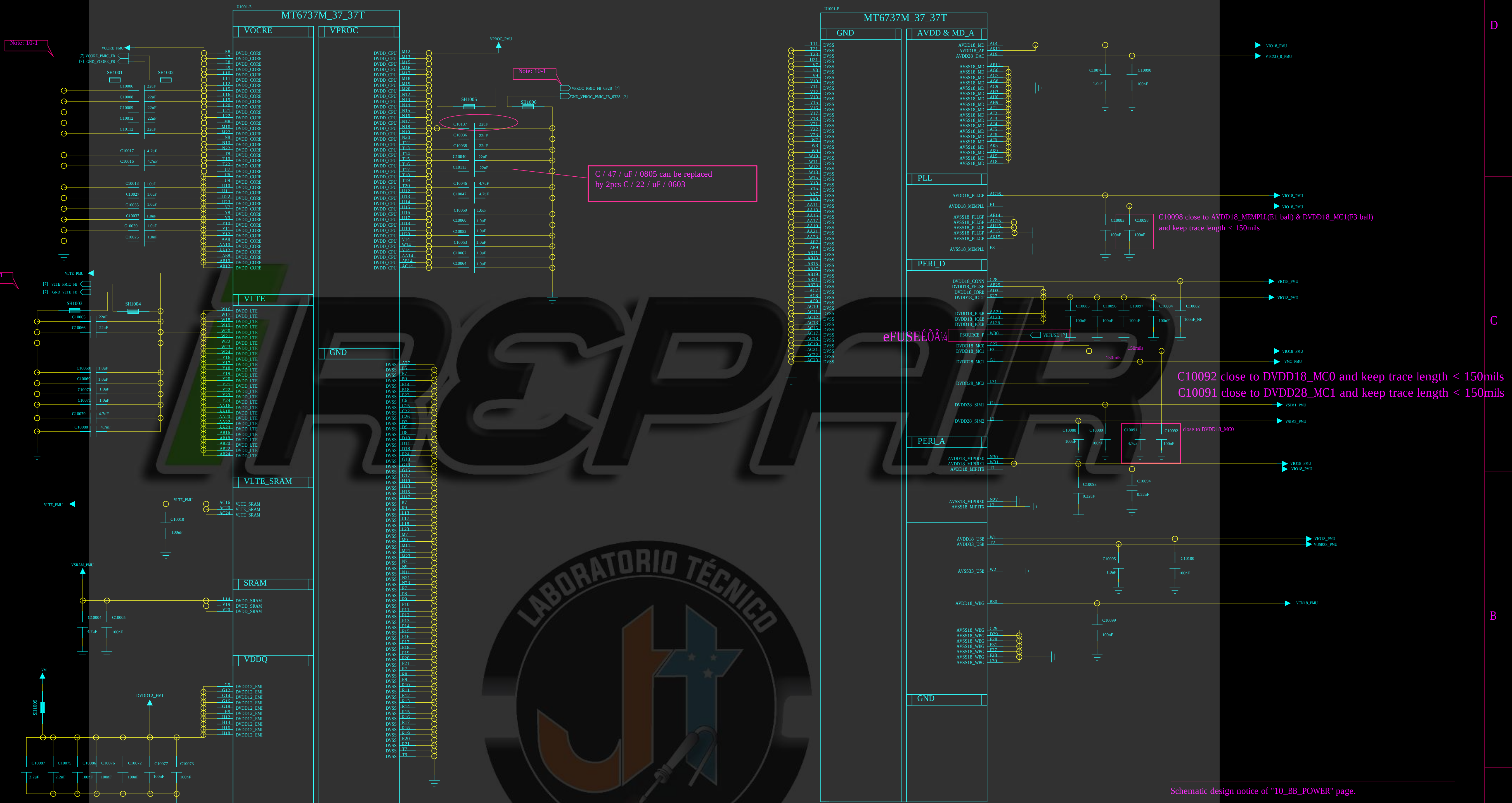
TITLE: XT1758

DRAWN: <Drawn By>	DATED: <Drawn Date>
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: <Released By>	DATED: <Release Date>

CODE: <Code>	SIZE: A1	DRAWING NO: <Drawing Number>	REV: 1_13
SCALE: <Scale>			SHEET 08 OF 21

MT6737M

REVISION RECORD			
UTR	ECO NO:	APPROVED:	DATE:



Schematic design notice of "10_BB_POWER" page.

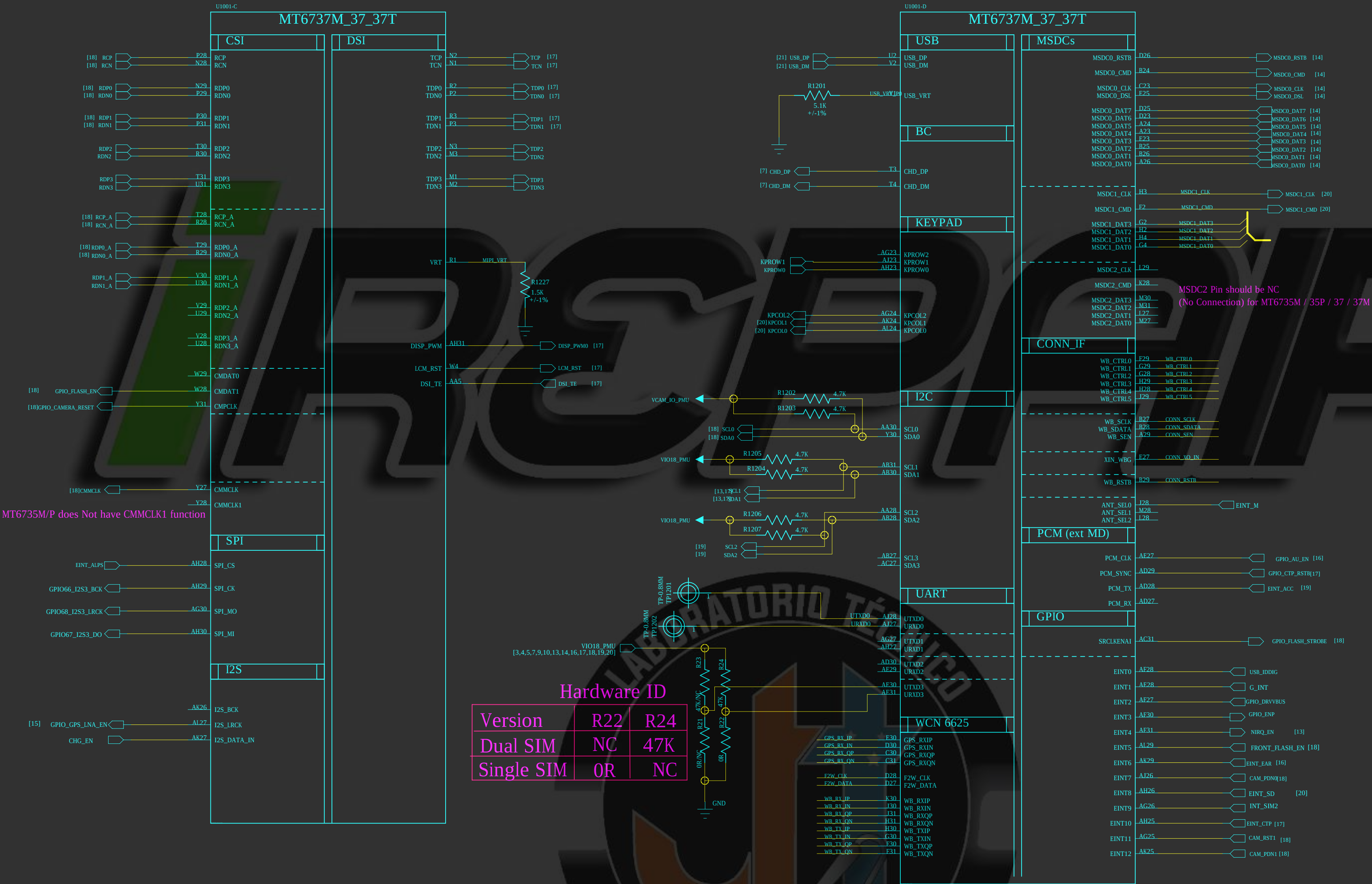
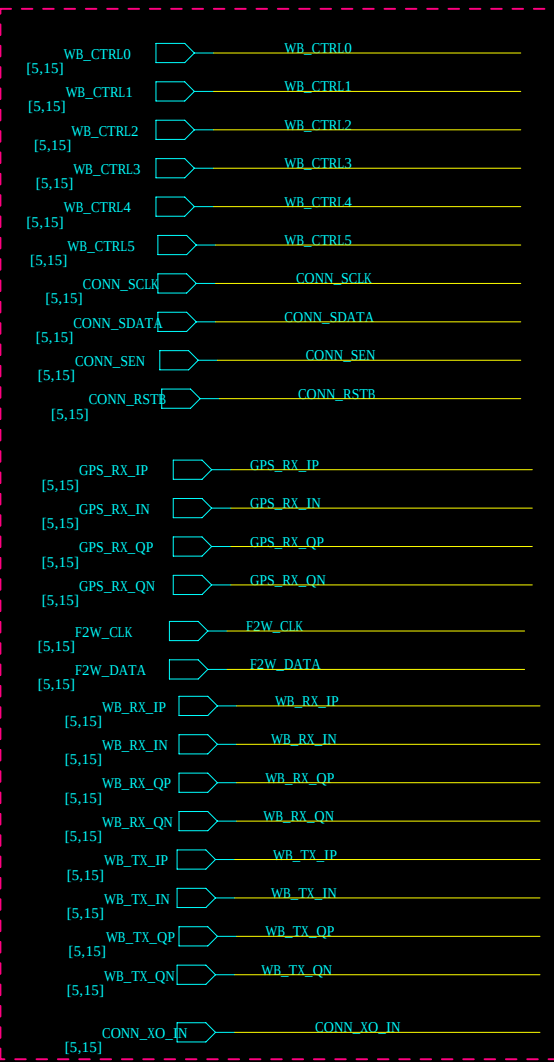
- Note 10-1: 4 mil GND trace with good shielding to PMIC (Differential)
- Note 10-3: FSOURCE_P(EFUSE)
(1)FSOURCE_P EFUSE power(VEFUSE) should be only for EFUSE usage(not share with other application)
(2)W/I EFUSE program, VEFUSE need 1uF bypass cap (pls refer to ¡§LDO output voltage/current table¡¨)
(3)W/O EFUSE program, VEFUSE bypass cap should be NC.

COMPANY:		Motorola			
TITLE:		XT1758			
DRAWN:	<Drawn By>	DATED:	<Drawn Date>	CODE:	SIZE:
CHECKED:	<Checked By>	DATED:	<Checked Date>	DRAWING NO:	REV:
QUALITY CONTROL:	<QC By>	DATED:	<QC Date>	<Code>	A1 <Drawing Number>
RELEASED:	<Released By>	DATED:	<Release Date>	SCALE:	<Scale>
SHEET: 21				21	

DRAWN: <Drawn By>	DATED: <Drawn Date>
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: <Released By>	DATED: <Release Date>

MT6737M

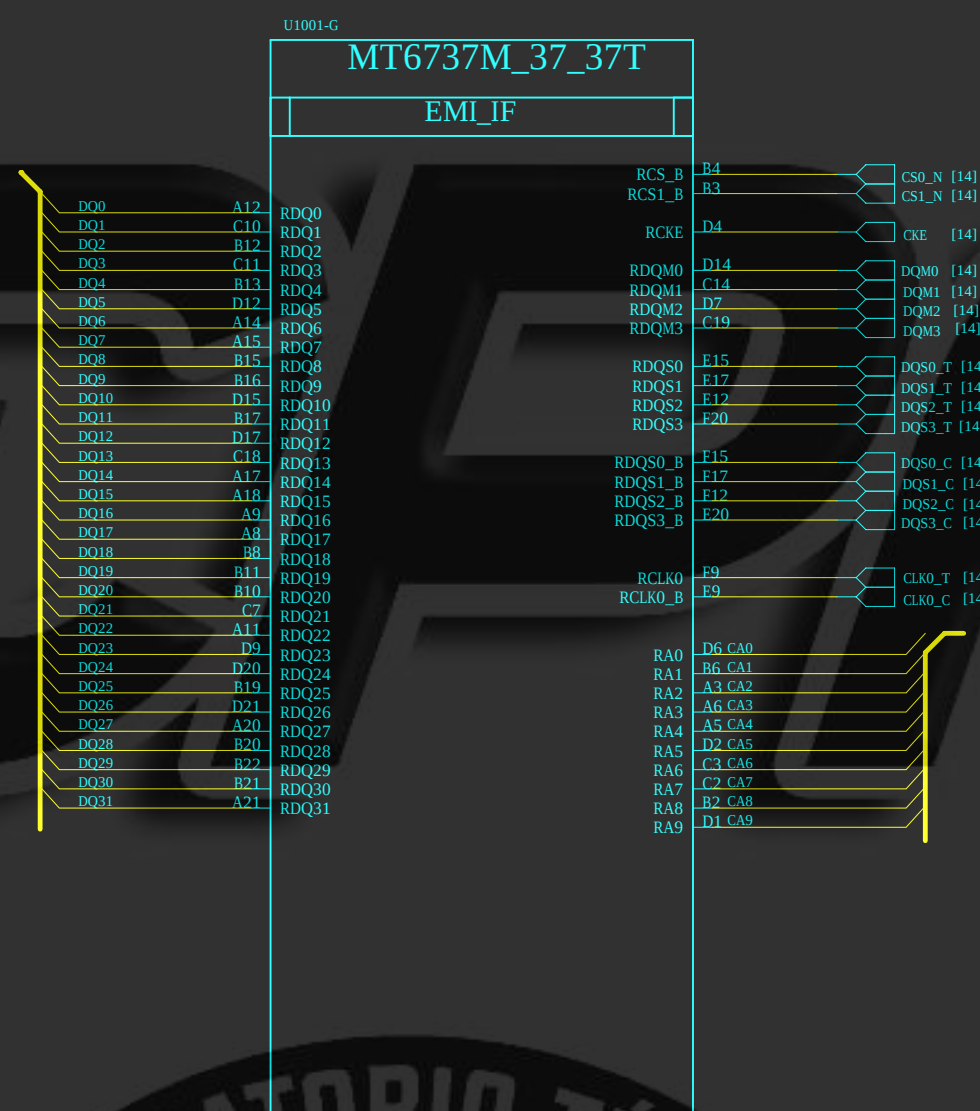
REVISION RECORD			
LTR	ECO NO.	APPROVED	DATE



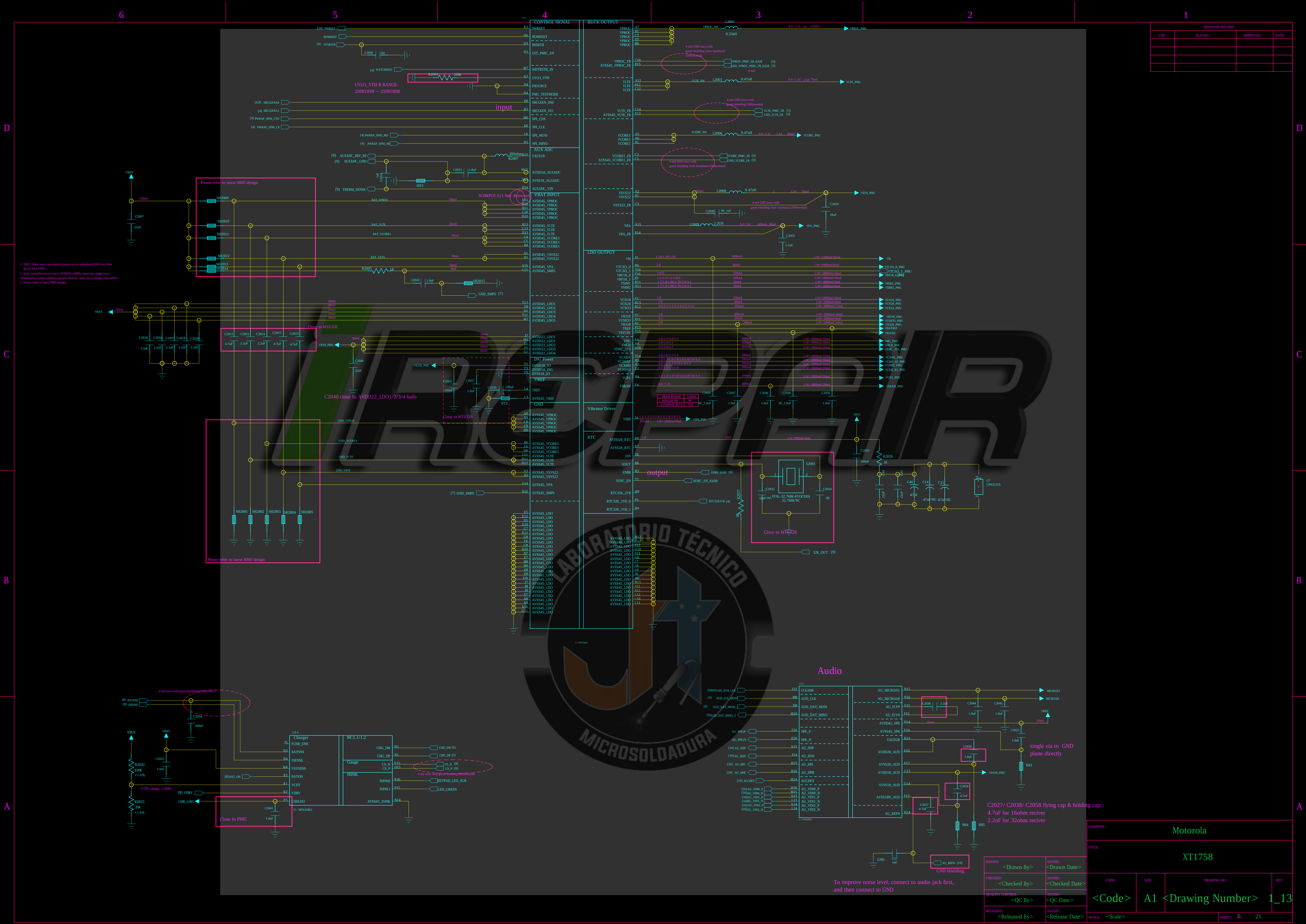
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TITLE: XT1758			
CODE: <Code>	SIZE: A1	DRAWING NO: <Drawing Number>	REV: 1_13
SCALE: <Scale>		SHEET: 6F	21

DRAWN: <Drawn By>	DATED: <Drawn Date>
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: <Released By>	DATED: <Release Date>

MT6737M



DRAWN: <Drawn By>		DATED: <Drawn Date>		A11/56			
CHECKED: <Checked By>		DATED: <Checked Date>		CODE:	SIZE:	DRAWING NO:	REV:
QUALITY CONTROL: <QC By>		DATED: <QC Date>		<Code>	A1	<Drawing Number>	1_13
RELEASED: <Released By>		DATED: <Release Date>		SCALE: <Scale>		SHEET: 66 21	



REVISION RECORD			
UTR	ECO NO.	APPROVED	DATE

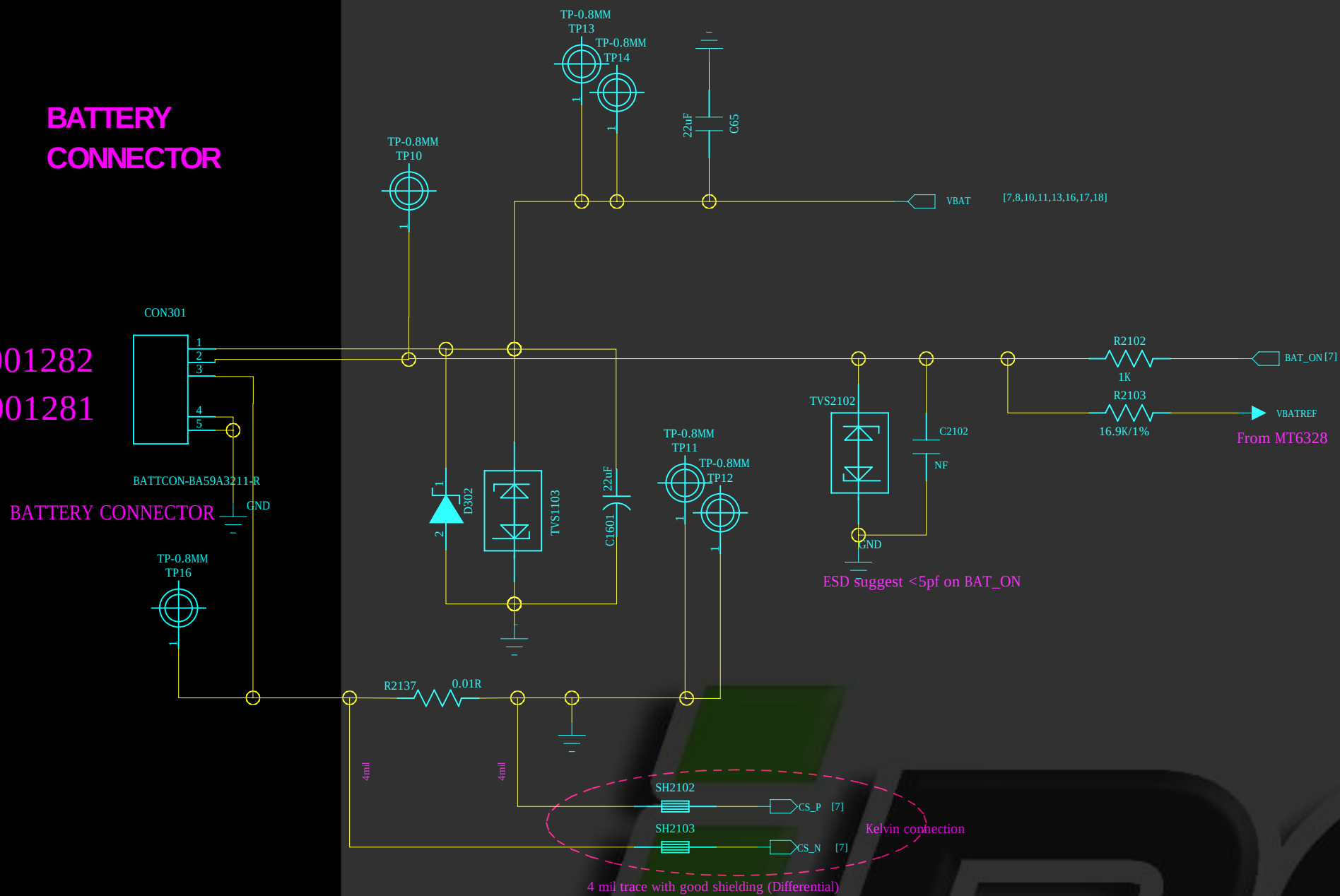
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TITLE: XT1758			
CODE:	SIZE:	DRAWING NO:	REV:
<Code>	A1	<Drawing Number>	1_13
SCALE: <Scale>	SHEET 8 of 21		

DRAWN: <Drawn By>	DATED: <Drawn Date>
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: <Released By>	DATED: <Release Date>

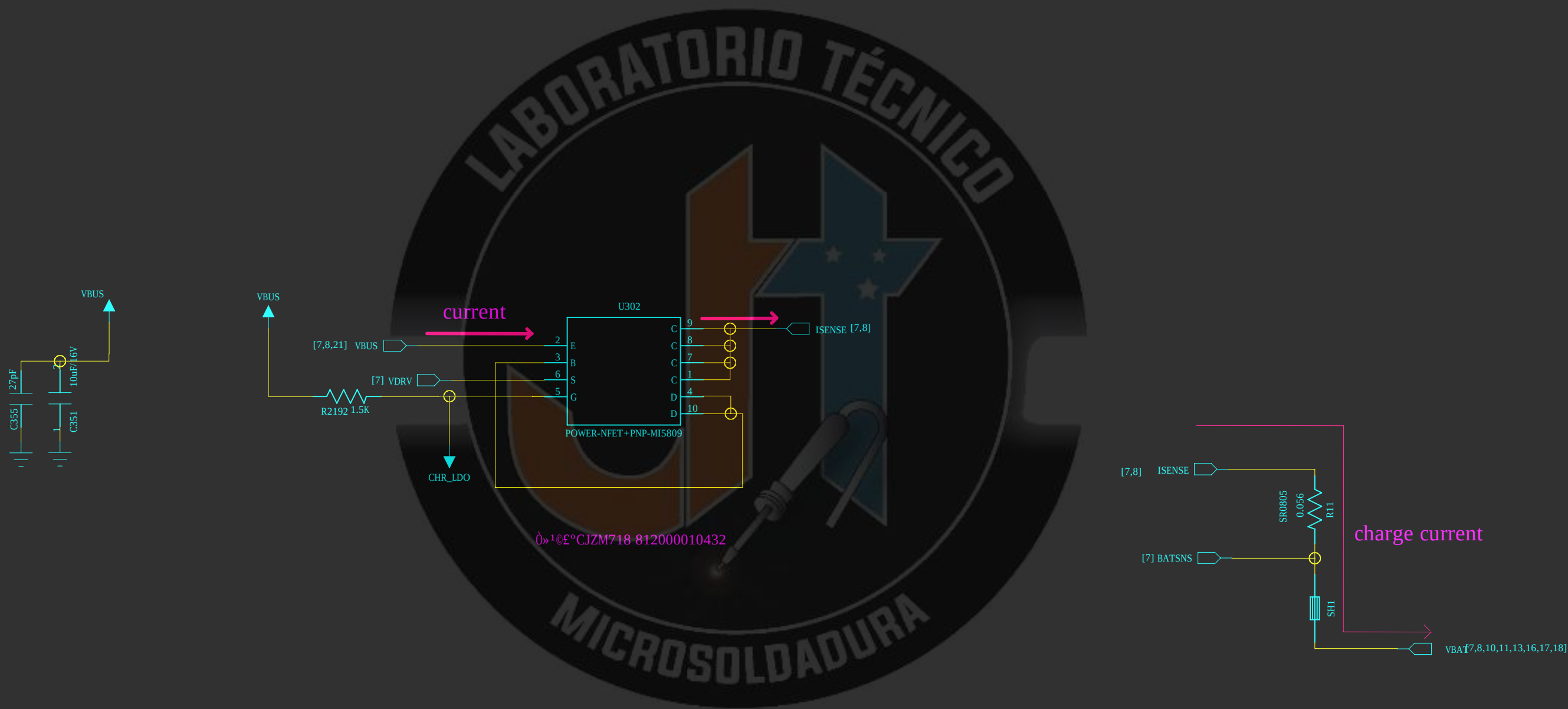
To improve noise level, connect to audio jack first, and then connect to GND

REVISION RECORD			
UTR	ECO NO.	APPROVED:	DATE:

BATTERY CONNECTOR



Charger



COMPANY: Motorola

TITLE: XT1758

CODE:	SIZE:	DRAWING NO:	REV:
<Code>	A1	<Drawing Number>	1_13

DRAWN: <Drawn By>	DATED: <Drawn Date>
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: <Released By>	DATED: <Release Date>

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

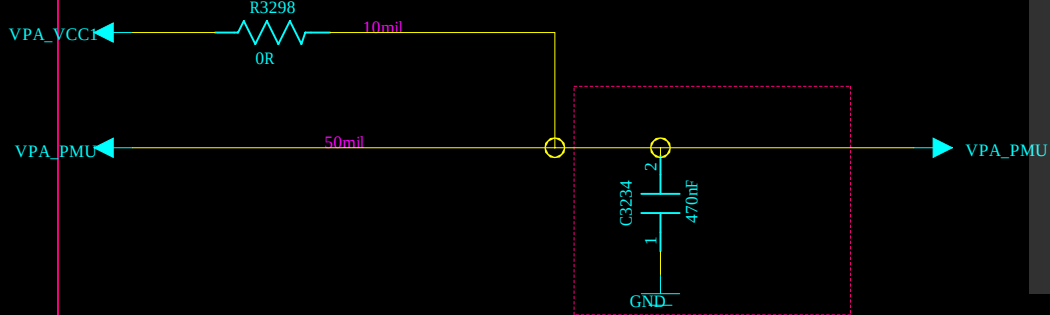
D

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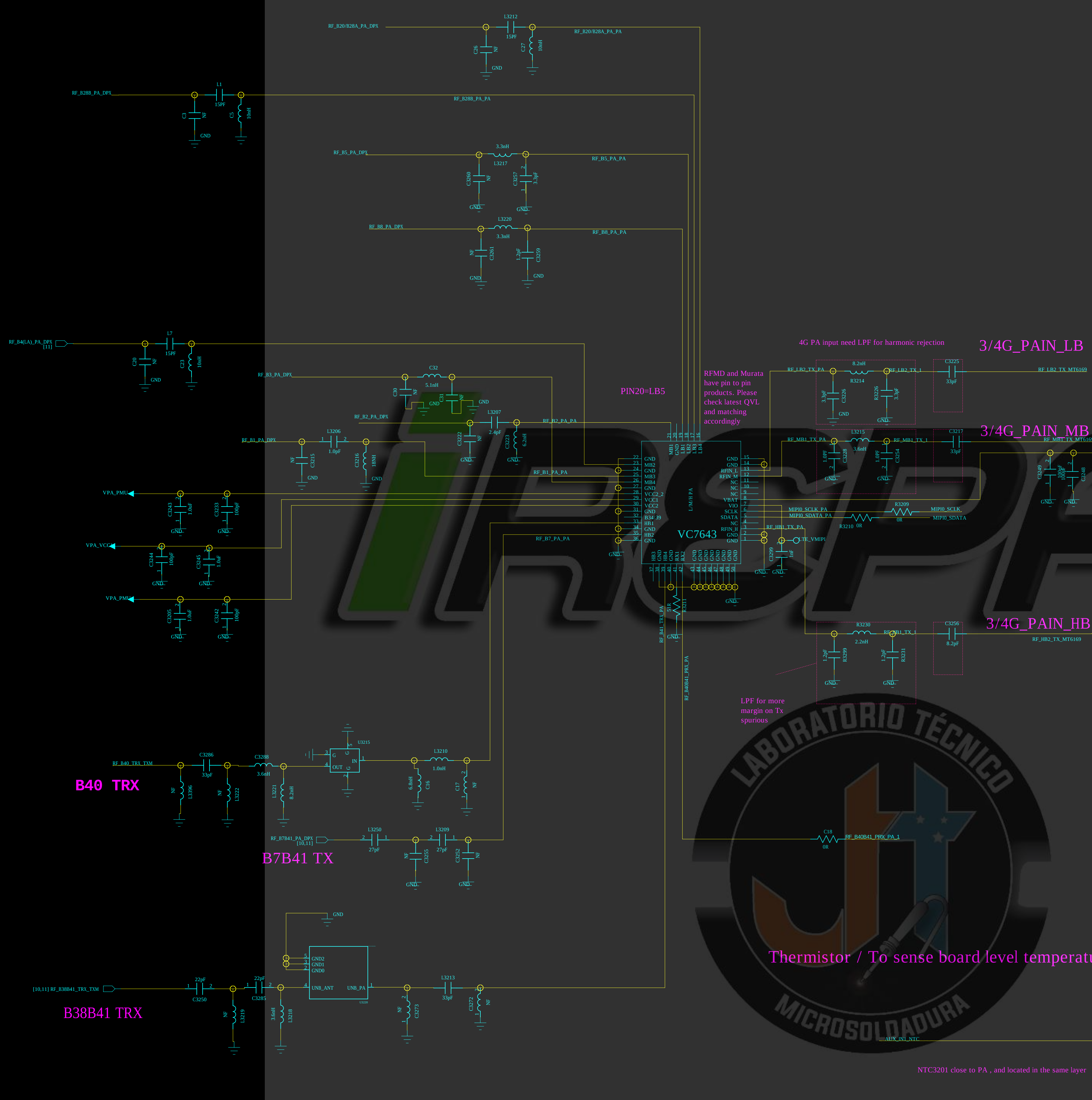
Power Net Connection



B40 TRX

B7B41 TX

B38B41 TRX



Thermistor / To sense board level temperature

NTC3201 close to PA , and located in the same layer

Off-Page Connector
(To RF front-end)

- [31] RF_BB2_TX_MTS169 [9,10]
- [31] RF_MB1_TX_MTS169 [9,10]
- [31] RF_LB2_TX_MTS169 [9,10]

- [33] RF_B5_PA_DPX [10,11]
- [33] RF_B3_PA_DPX [10,11]
- [33] RF_B2_PA_DPX [10,11]
- [33] RF_B1_PA_DPX [10,11]
- [33] RF_B7B1_TX_MTS169 [10,11]
- [33] RF_B7B2_TX_MTS169 [10,11]
- [33] RF_B8_PA_DPX [10,11]

- [99] AUX_IN1_NTC [4,10]

- MIPID0_SDATA [99]
- MIPID0_SCLK [99]

- VPA_PMU [99]
- VPA_PMU [99]

- [10,11] RF_B20/B28A_PA_DPX [10,11]
- [10,11] RF_B28B_PA_DPX [10,11]
- [11] RF_B12_PA_DPX [10,11]
- [11] RF_B14/LANA1_PA_DPX [10,11]
- [10,11] RF_B40_TX_TXM [10,11]
- [10,11] RF_B3841_TX_TXM [10,11]

COMPANY: Motorola

TITLE: XT1758

CODE: <Code> SIZE: A1 DRAWING NO.: <Drawing Number> REV: 1_13

SCALE: <Scale> SHEET: 20 21

DRAWN:	<Drawn By>	DATED:	<Drawn Date>
CHECKED:	<Checked By>	DATED:	<Checked Date>
QUALITY CONTROL:	<QC By>	DATED:	<QC Date>
RELEASED:	<Released By>	DATED:	<Release Date>



REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

Connect to AP

DQ[0..31]	DQ[0..31]
CA[0..9]	CA[0..9]
CS0_N	CS0_N
CS1_N	CS1_N
CKE	CKE
DQM0	DQM0
DQM1	DQM1
DQM2	DQM2
DQM3	DQM3
DQS0_C	DQS0_C
DQS1_C	DQS1_C
DQS2_C	DQS2_C
DQS3_C	DQS3_C
DQS0_T	DQS0_T
DQS1_T	DQS1_T
DQS2_T	DQS2_T
DQS3_T	DQS3_T
CLK0_T	CLK0_T
CLK0_C	CLK0_C
VREF_CA	VREF_CA
VREF_DQ	VREF_DQ

MSDC0_RSTB	MSDC0_RSTB
MSDC0_CMD	MSDC0_CMD
MSDC0_CLK	MSDC0_CLK
MSDC0_DSL	MSDC0_DSL
MSDC0_DAT0	MSDC0_DAT0
MSDC0_DAT1	MSDC0_DAT1
MSDC0_DAT2	MSDC0_DAT2
MSDC0_DAT3	MSDC0_DAT3
MSDC0_DAT4	MSDC0_DAT4
MSDC0_DAT5	MSDC0_DAT5
MSDC0_DAT6	MSDC0_DAT6
MSDC0_DAT7	MSDC0_DAT7

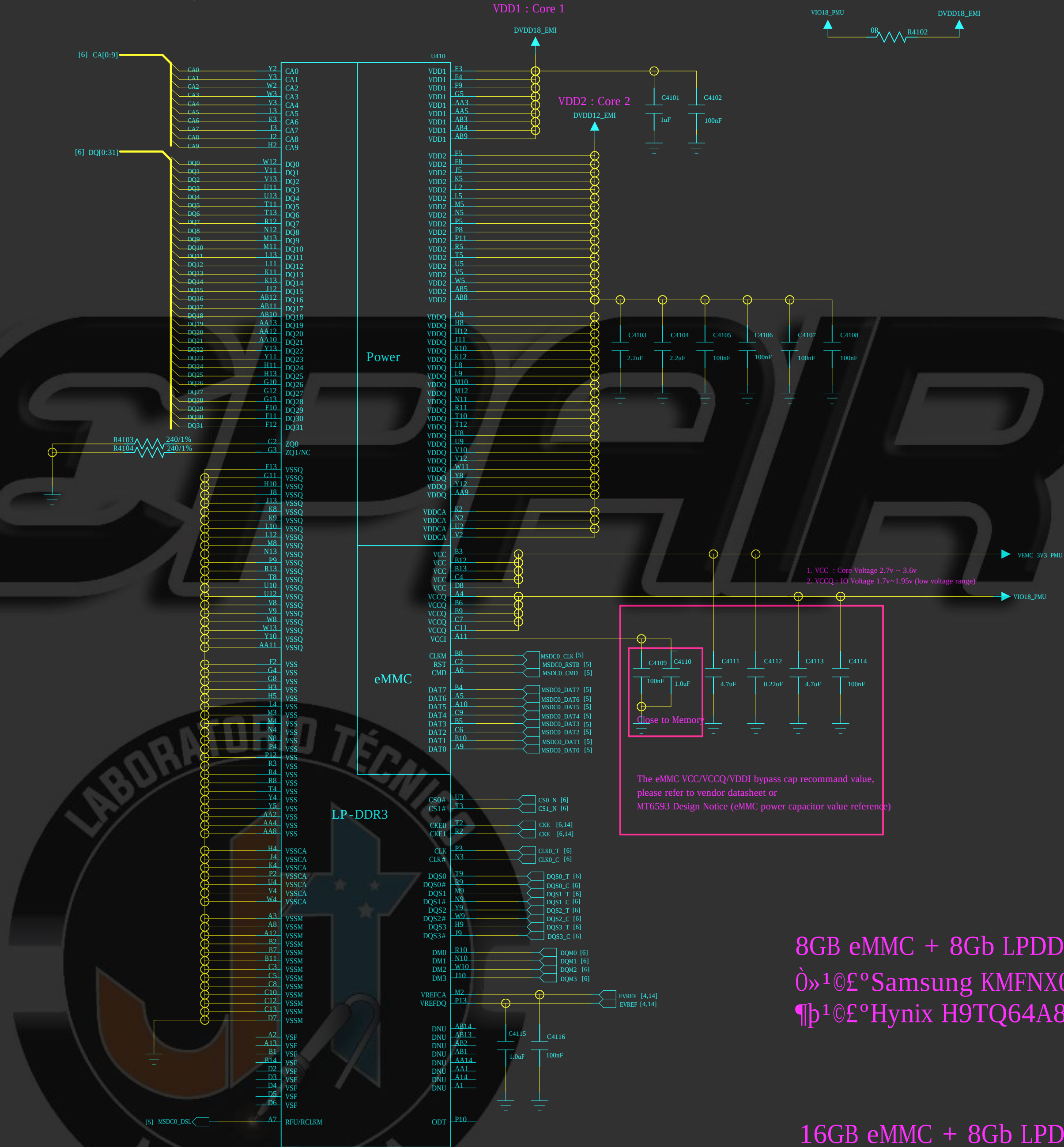
Power I/F

VIO18_PMU	VIO18_PMU
VEMC_3V3_PMU	VEMC_3V3_PMU
DVDD12_EMI	DVDD12_EMI

eMMC+LPDDR3

221 Ball, 0.5mm pitch

VDD1=1.8V
VDD2=1.20V
VDDCA=1.2V
VDDQ= 1.20V



8GB eMMC + 8Gb LPDDR3

0»¹@£°Samsung KMFNX0012M-B214 806700000431

¶p¹@£°Hynix H9TQ64A8GTBCUR-KUM

16GB eMMC + 8Gb LPDDR3

0»¹@£°Samsung KMFE10012M-B214 806700000291

¶p¹@£°Hynix H9TQ17A8GTACUR-KUM

COMPANY: Motorola

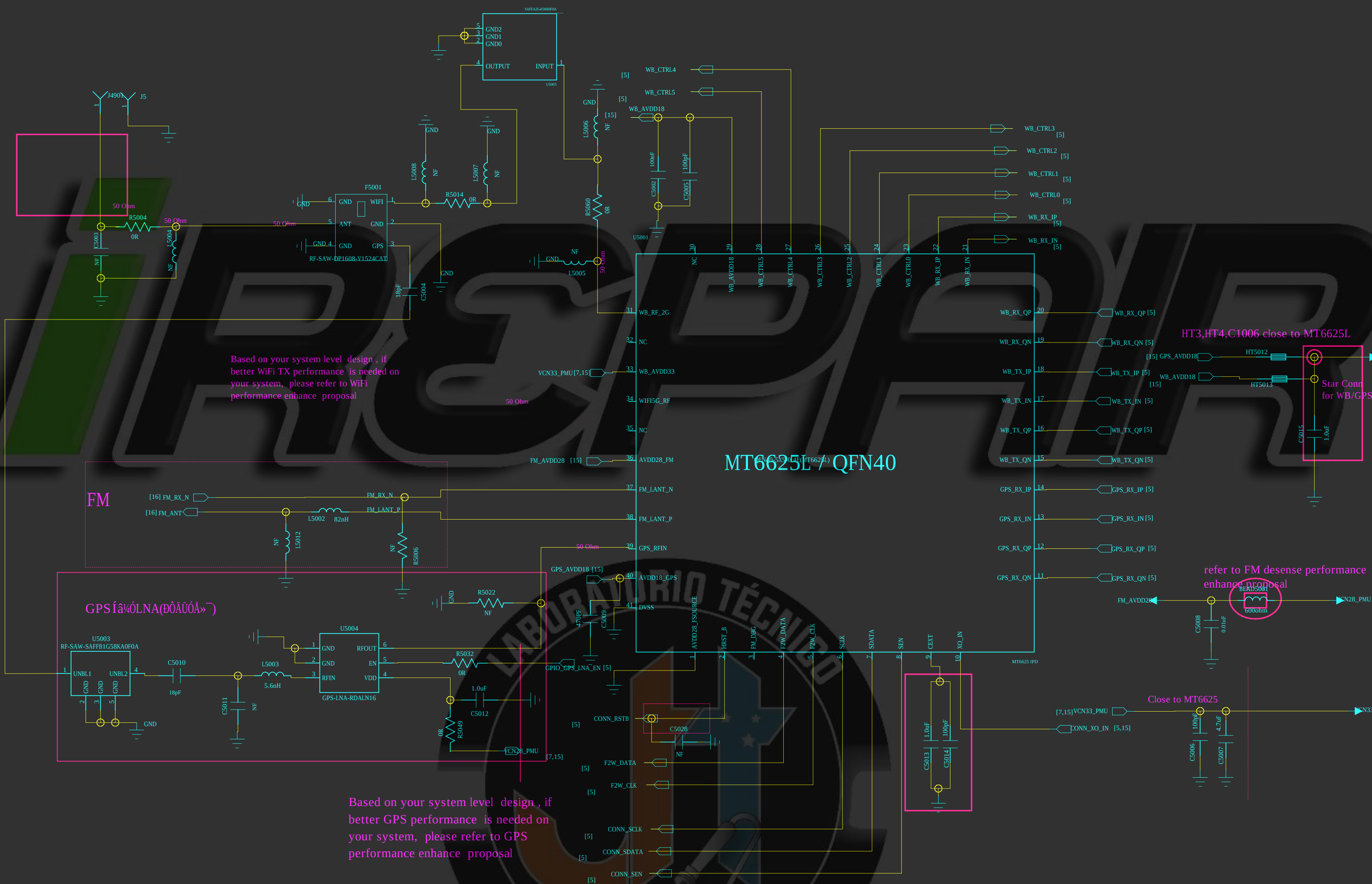
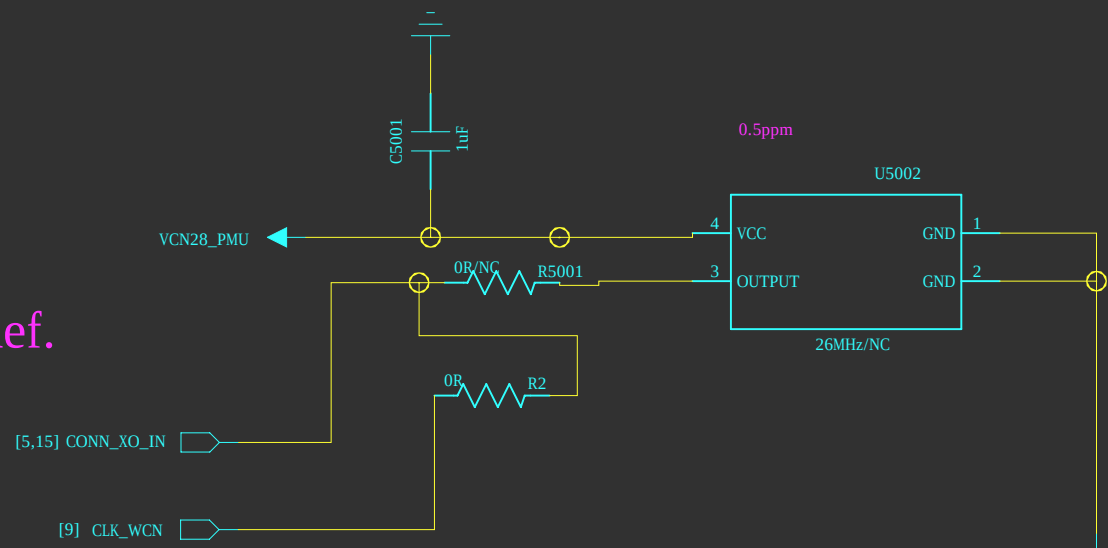
TITLE: XT1758

CODE:	SIZE:	DRAWING NO:	REV:
<Code>	A1	<Drawing Number>	1_13
SHEET 84		21	

DRAWN:	DATED:
<Drawn By>	<Drawn Date>
CHECKED:	DATED:
<Checked By>	<Checked Date>
QUALITY CONTROL:	DATED:
<QC By>	<QC Date>
RELEASED:	DATED:
<Released By>	<Release Date>

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

WIFI/BT/GPS Single ANT Ref.



Based on your system level design , if better WIFI TX performance is needed on your system, please refer to WIFI performance enhance proposal

MT6625L / QFN40

HT3,HT4,C1006 close to MT6625L

Star Conn for WB/GPS/WBG 1V8

refer to FM desense performance enhance proposal

Close to MT6625

Based on your system level design , if better GPS performance is needed on your system, please refer to GPS performance enhance proposal

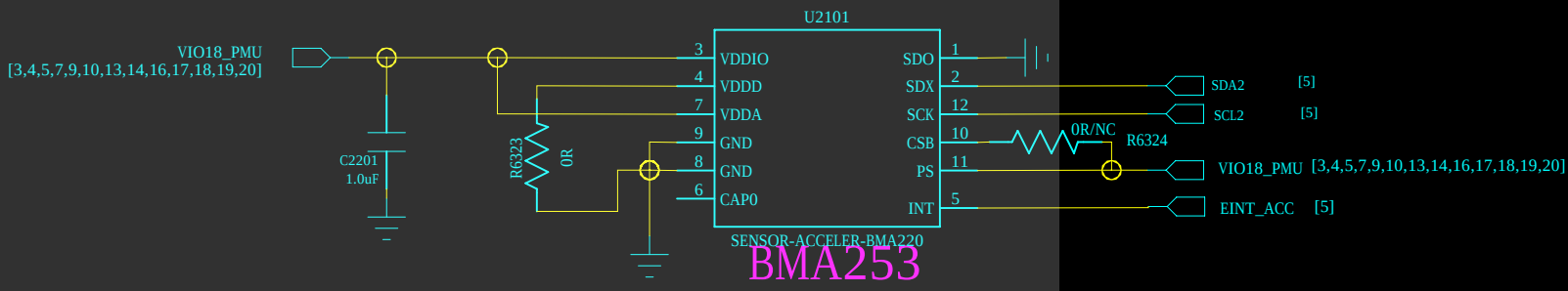
COMPANY: Motorola			
TITLE: XT1758			
CODE: <Code>	SIZE: A1	DRAWING NO: <Drawing Number>	REV: 1_13
SHEET: 85		21	

DRAWN: <Drawn By>	DATED: <Drawn Date>
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: <Released By>	DATED: <Release Date>

M-Sensor

G-Sensor

REVISION RECORD			
UTR	ECO NO.	APPROVED:	DATE:



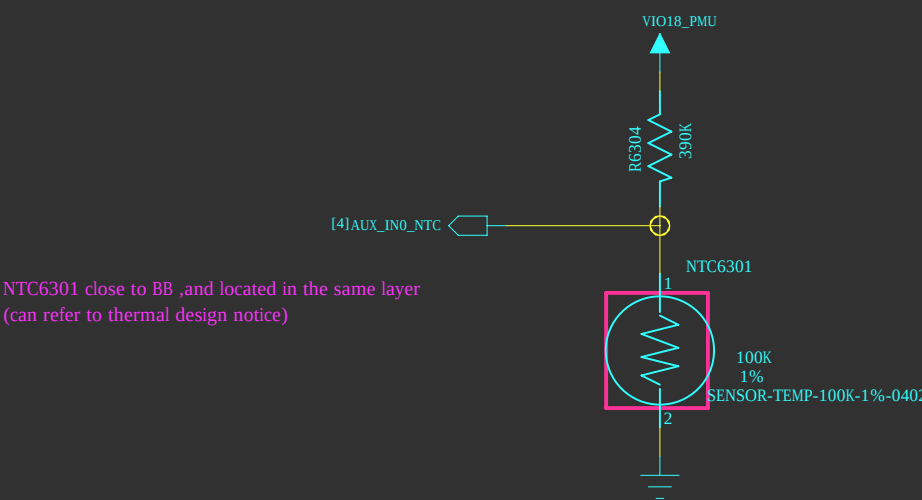
BMA: R6323 0 R6324 NF
BMC156: R6323 0 R6324 0
KXT12-1000: R6323 NF R6324 0

0»¹©£°821100000031
¶þ¹©£°821100000041

Accerometer

ALS+PS+IR

Thermistor



Thermistor / To sense board level temperature

COMPANY: Motorola			
TITLE: XT1758			
CODE: <Code>	SIZE: A1	DRAWING NO: <Drawing Number>	REV: 1_13
SCALE: <Scale>			SHEET 00 21

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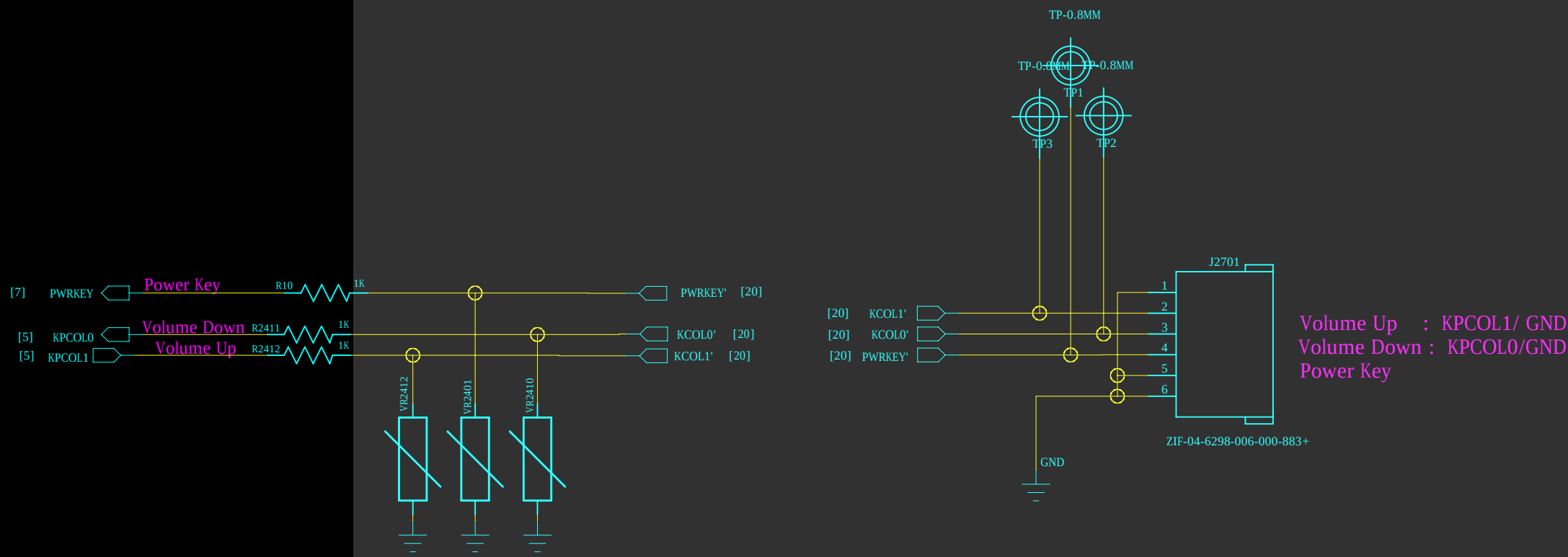
D

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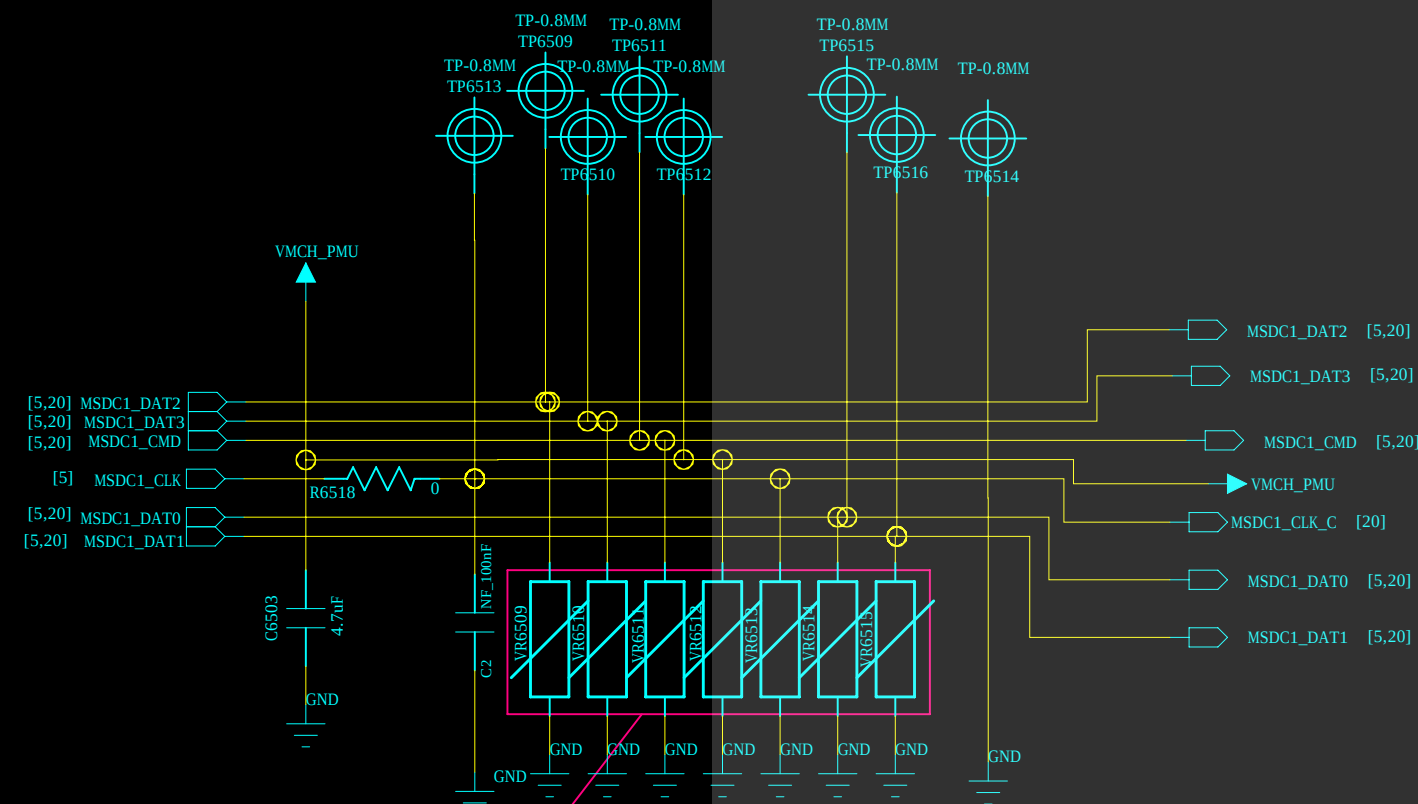
REVISION RECORD			
UTR	ECO NO.	APPROVED:	DATE:



0»¹@£°813200000193

¶p¹@£°813200000192

REPAIR

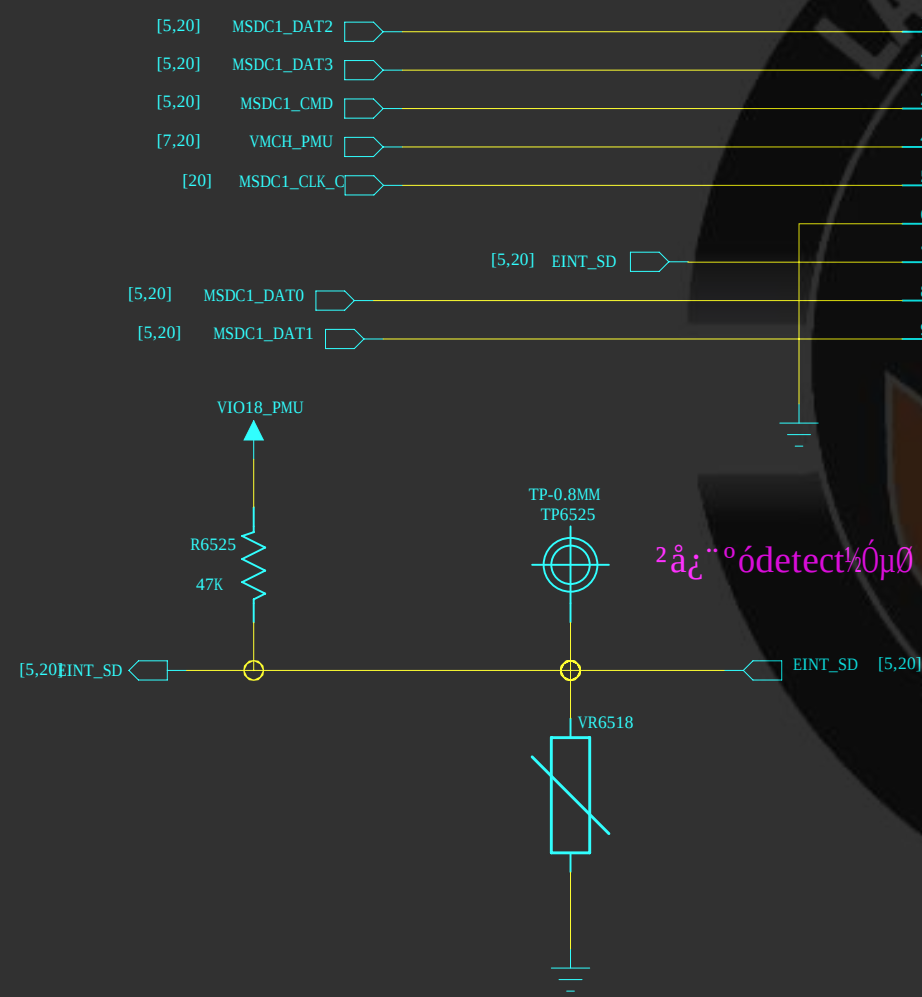


Note: 40-1

Based on your system level design , if better ESD/desense performance is needed on your system.

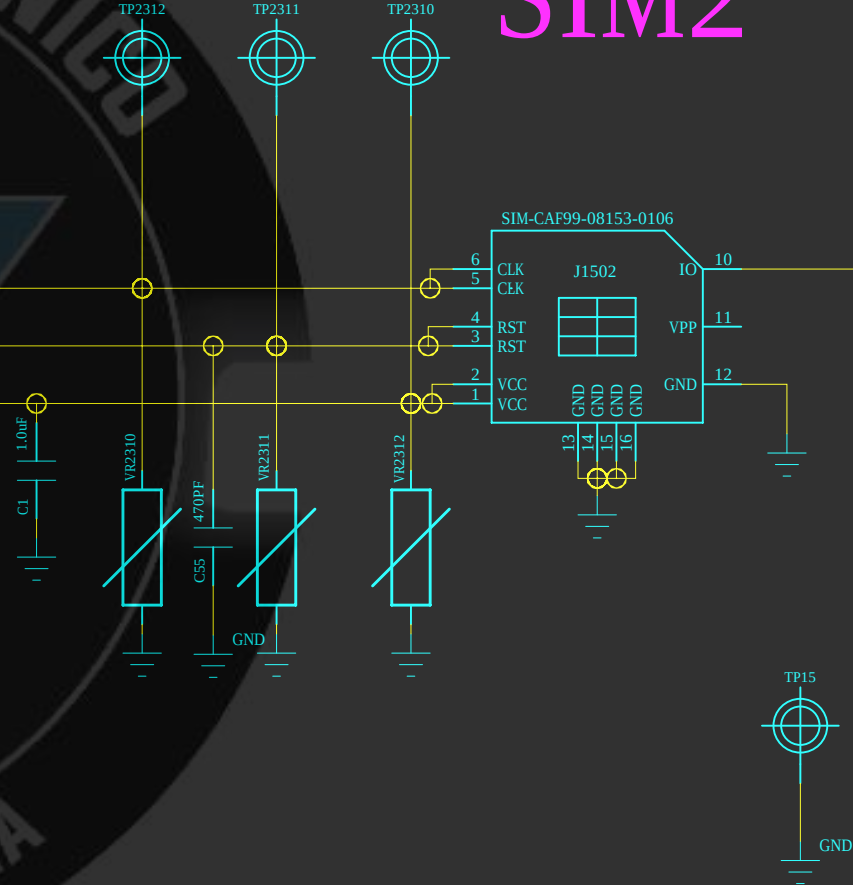
0»¹@£°813320000041

¶p¹@£°813320000042

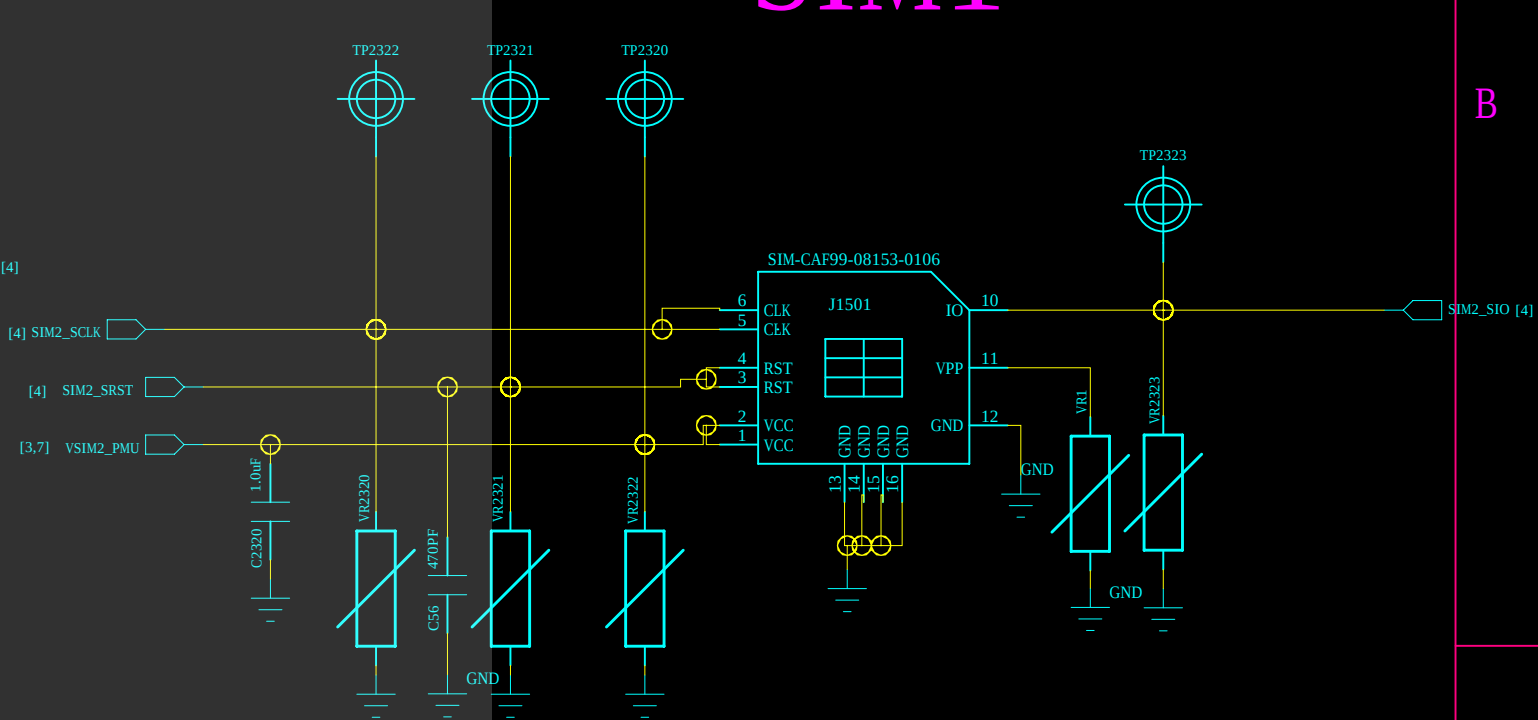


2â¿""°ódetect%0µ0

SIM2



SIM1



0»¹@£°813310000221

¶p¹@£°813310000222

Schematic design notice of "40_MEMORY_SD Card" page.

Note 40-1: The equivalent capacitance of MSDC ESD protection device must be <=10pF.
But for NFC app. equivalent capacitance of MSDC_NFC_SWPIO and MSDC_NFC_VCCSWP should <=0.5pF.

COMPANY: Motorola

TITLE: XT1758

DRAWN:	<Drawn By>	DATED:	<Drawn Date>
CHECKED:	<Checked By>	DATED:	<Checked Date>
QUALITY CONTROL:	<QC By>	DATED:	<QC Date>
RELEASED:	<Released By>	DATED:	<Release Date>

CODE:	SIZE:	DRAWING NO:	REV:
<Code>	A1	<Drawing Number>	1_13
SCALE:	<Scale>	SHEET	20 21

CODE:	SIZE:	DRAWING NO:	REV:
<Code>	A1	<Drawing Number>	1_13
SCALE: <Scale>		SHEET: 01	21

DRAWN: <Drawn By>	DATED: <Drawn Date>
CHECKED: <Checked By>	DATED: <Checked Date>
QUALITY CONTROL: <QC By>	DATED: <QC Date>
RELEASED: <Released By>	DATED: <Release Date>